Climate Data Management System (CDMS) Functional and Non-functional Requirements

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Chapter 1. Scope

1.1. Introduction

1.1.1. Purpose and Relationship to WMO-No. 1131

This document defines the minimal functional and non-functional requirements to support Members in establishing climate data management systems or assessing the compliance of existing systems with the *Climate Data Management System Specifications* (WMO-No. 1131, 2025 Edition).

WMO-No. 1131 provides a comprehensive framework for Climate Data Management Systems (CDMS), describing key components, governance principles, and operational practices using the classifications "Required", "Recommended", and "Optional". However, Members may require more concrete technical requirements to support implementation.

This document translates the framework outlined in WMO-No. 1131 into a set of concrete functional and non-functional requirements. These requirements represent a subset of the full specifications in WMO-No. 1131 and serve as a practical and technical specification, focusing on capabilities essential for managing observational data from stations, including station, sensor, and provenance metadata.

The functional requirements are expressed as requirements classes following the OGC Modular Specification approach.

This specification aims to:

- **Assist Developers and Integrators**: Provide practical, testable requirements and detailed technical considerations for the design, development, and integration of robust and compliant CDMS components and systems.
- **Empower Users and Assessors**: Establish a comprehensive set of conformance criteria and common understanding to objectively assess the capabilities and compliance of existing, or proposed, CDMS implementations with the essential functionalities described in WMO-No. 1131.
- **Enable Interoperability**: Define standardized interfaces, data formats, and protocols to facilitate integration between CDMS implementations and external systems within the climate data ecosystem.

This specification is *not intended* as a list of requirements that WMO Members SHALL or SHOULD follow.

1.1.2. System Context and Boundaries

Figure 1 illustrates the architectural landscape and positioning of a Climate Data Management System within the broader weather, climate, and other environmental data value chain (hereafter, climate data for conciseness). Climate data typically flows from field sensors and observing stations through national Data Collection and Processing Systems (DCPS), and subsequently to various downstream applications and services.

Key downstream systems include:

- WIS 2.0 nodes for international data exchange
- Operational meteorological systems (Numerical Weather Prediction, data assimilation, forecasting systems)
- Climate Data Management Systems for quality control, homogenization, and archival
- Regional Climate Centres for regional climate monitoring and services

- Long-term preservation archives and public-facing data portals

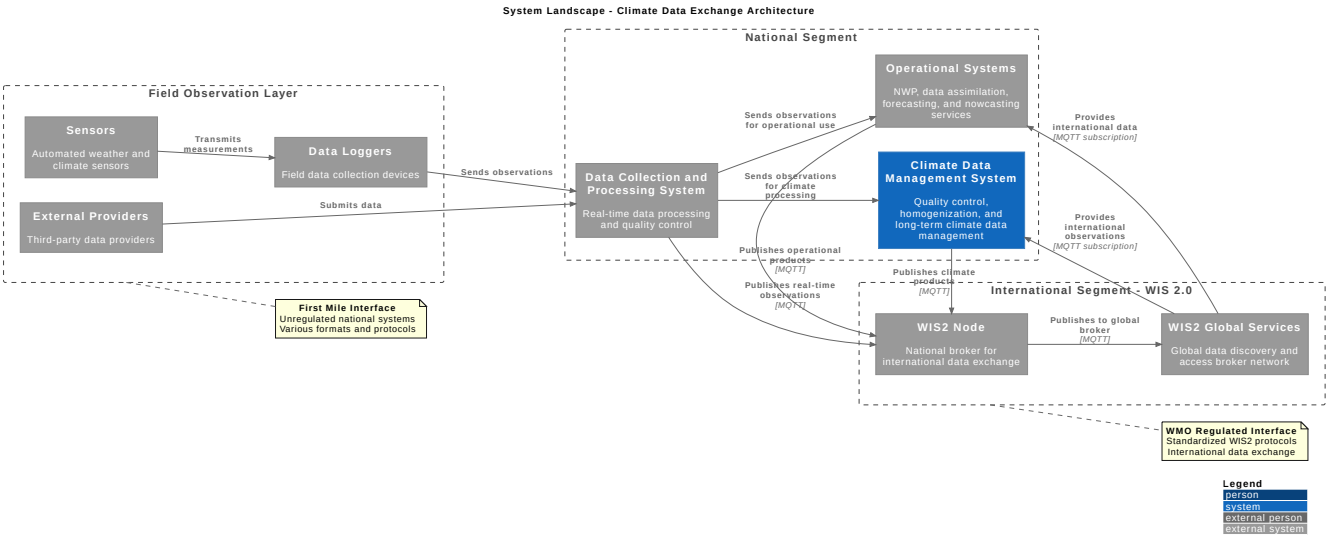

Figure 1. System landscape showing the context of Climate Data Management Systems within the data exchange architecture

Scope Boundaries:

While some operational implementations may integrate multiple components of the data flow—for example, combining DCPS and CDMS functionality within a single system—this specification focuses exclusively on the Climate Data Management System component and its specific responsibilities.

The following are explicitly OUT OF SCOPE for this specification:

- Field observation collection (sensors, data loggers, first-mile infrastructure)
- Real-time data collection and processing systems (DCPS)
- Time-critical data exchange via WIS 2.0 for operational meteorology
- Numerical Weather Prediction and forecasting systems
- General-purpose meteorological databases not specifically designed for climate data
- Operational quality control for real-time applications

The following are IN SCOPE:

- Quality control procedures specific to climate data requirements with emphasis on long-term consistency
- Detection and correction of inhomogeneities in climate time series (homogenization)
- Climate data rescue, including digitization of historical paper records
- Station metadata management with emphasis on long-term consistency and provenance
- Archival and preservation of climate datasets with appropriate documentation
- Production, publication and distribution of quality-controlled climate data products
- Integration interfaces with WIS 2.0, national networks, and international archives
- Data lineage tracking and provenance management

This specification recognizes that CDMS implementations may exist within national meteorological services (National Segment) and contribute to regional and international climate services (Regional/International Segments) as shown in Figure 1. Requirements are designed to support both contexts while maintaining

flexibility for local operational arrangements.

1.2. System Architecture

1.2.1. System Context (Level 1)

Figure 2 illustrates the system context for a Climate Data Management System, showing the primary users, the CDMS boundary, and its interactions with external systems. The diagram identifies key integration points including:

- **Data sources**: The national Data Collection and Processing System (DCPS), and WIS 2.0 Global Services (via national WIS2 Node).
- **Users**: Climate scientists, data managers, and system administrators
- **Downstream systems**: Long-term archives, climate data stores, and climate analysis platforms
- **Support systems**: Identity management and operations monitoring

While these external systems and their interactions are shown for context, their internal specifications are outside the scope of this document. This specification focuses exclusively on the requirements for the CDMS component itself and its standardized interfaces with these external systems.

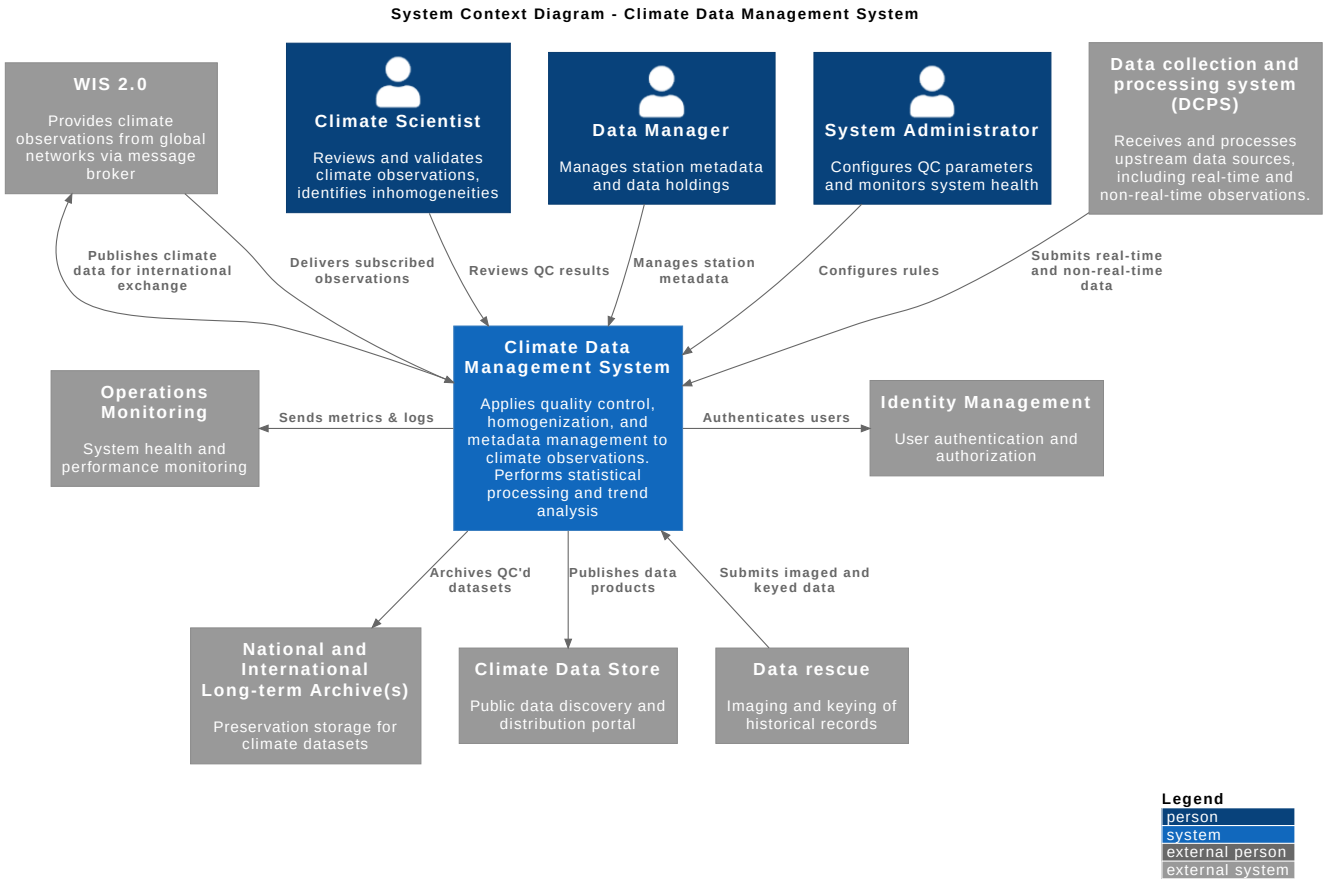

Figure 2. CDMS system context diagram showing users and external system interfaces

1.2.2. Container Architecture (Level 2)

Figure 3 provides a high-level view of the major functional components and data stores within the Climate Data Management System. This diagram is intentionally technology-agnostic, allowing implementers flexibility in architectural and technology choices while conforming to the functional requirements of this specification.

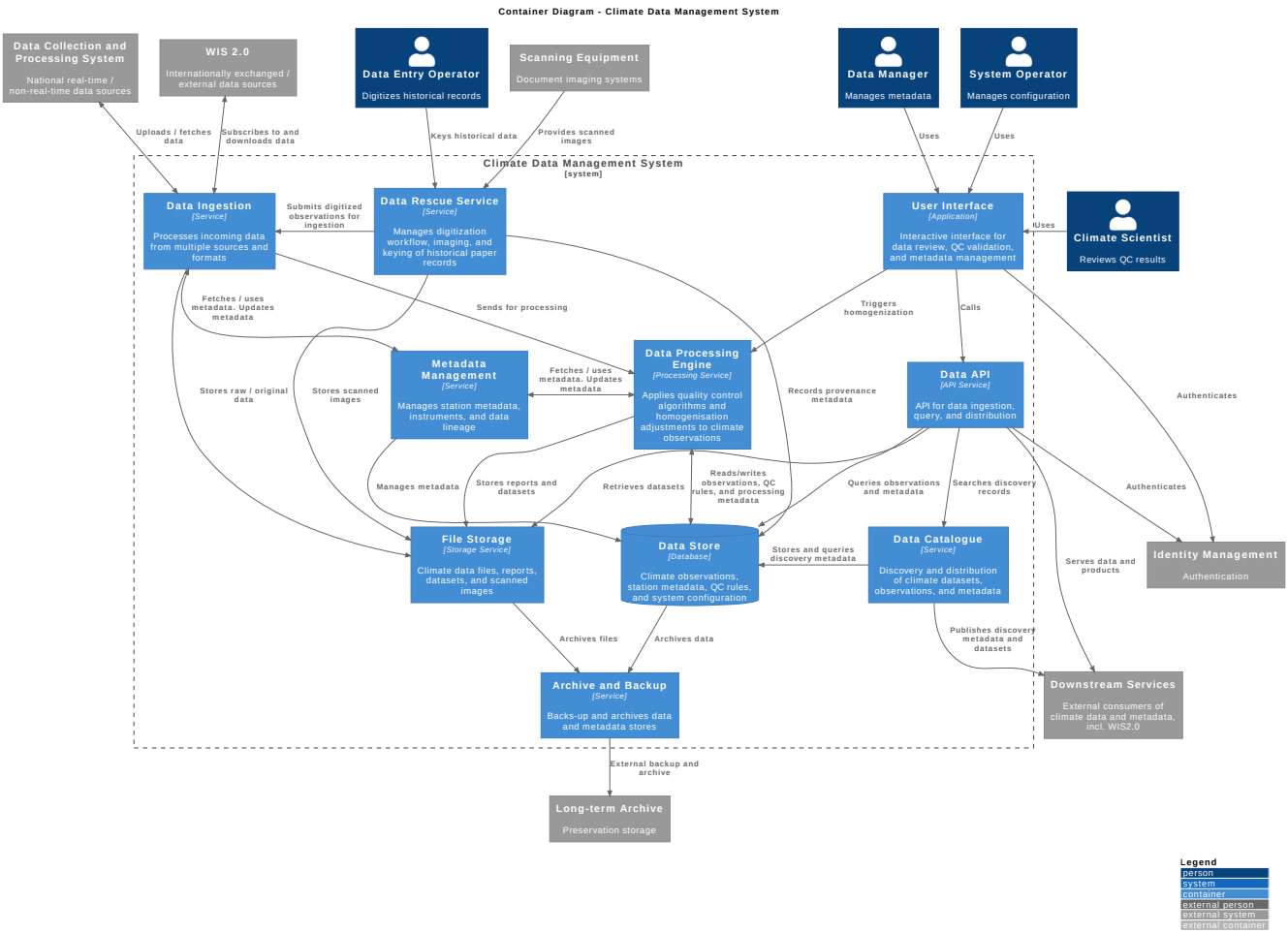

Figure 3. Container diagram showing major functional components and data stores

Key Containers:

- **User Interface**: Provides interactive capabilities for climate scientists, data managers, and system administrators to review quality-controlled data, validate observations, manage station metadata, configure system parameters, and monitor system operations. This may be implemented as a web application, desktop application, or a combination of both.
- **Data API**: Exposes standardized interfaces for data ingestion from external sources, data query and retrieval by users and downstream systems, and publication of climate data products. The API should support international standards for climate data exchange including OGC API standards.
- **Data Ingestion**: Processes incoming observations from the national DCPS (national real-time and non-real-time data) and from WIS 2.0 (international observations), as well as digitized historical observations submitted by the Data Rescue Service. Handles multiple data formats and protocols, performing initial validation and normalization.
- **Data Processing Engine**: Applies quality control algorithms to climate observations including range checks, temporal consistency checks, spatial consistency checks, and climatological validation. Also performs statistical analysis to detect and adjust for non-climatic discontinuities in observation time series (homogenization), producing both adjusted data and comprehensive documentation of the process.
- **Metadata Management**: Maintains comprehensive station and element metadata including location information, instrumentation history, observing practices, element definitions (units, codes, and controlled vocabularies), and data lineage. Supports both data ingestion validation and quality control processing, and ensures compliance with international metadata standards.
- **Data Rescue Service**: Manages the digitization workflow for historical climate records, including coordination of document imaging, data entry from paper records, and quality assurance of digitized

observations. Preserves both the scanned images and the keyed data with full provenance information. Digitized observations flow through the same Data Processing Engine as contemporary observations to ensure consistent quality standards.

- **Data Catalogue**: Maintains persistent discovery metadata records describing available datasets, observations, and metadata holdings. Supports search and discovery via the Data API and publishes discovery metadata to downstream services.
- **Archive and Backup**: Manages scheduled backup and archival of the Data Store and File Storage contents. Submits datasets to external long-term preservation archives and maintains backup copies to support disaster recovery.

Data Stores:

- **Data Store**: Persistent storage for climate observations with their associated quality flags and processing history, station and element metadata (including units, codes, and controlled vocabularies), quality control rules and thresholds, system configuration, homogenization parameters, and data lineage information. Maintains historical records of all metadata changes to support long-term climate analysis.
- **File Storage**: Manages climate-standard file formats (such as NetCDF-CF), quality control reports, homogenization analysis outputs, digitized document images, and archived datasets. Supports versioning and long-term preservation requirements.

External System Integration:

The system integrates with external systems at well-defined conformance boundaries where standardized protocols and data formats apply:

- **WIS 2.0**: International data exchange using MQTT messaging and standardized data formats for both consuming international observations and publishing national climate products.
- **National Data Collection and Processing System**: Real-time and non-real-time data sources and processing, including connection to data loggers, sensors and other external data sources.
- **Long-term Preservation Archives**: Climate dataset archival using standard formats.
- **Downstream Services**: External consumers of climate data and metadata, including climate data stores, analysis platforms, and WIS 2.0 nodes, accessed via the Data API and Data Catalogue.
- **Identity Management Systems**: Enterprise authentication and authorization using standard protocols (LDAP, OAuth 2.0, SAML).

This architectural decomposition supports the modular requirements structure of this specification, where different containers may be subject to different requirements classes based on their functional responsibilities.

1.3. Document Structure and Organization

To facilitate clear understanding and practical application, this document is structured into the following key sections:

- **Functional Requirements**: Details what a CDMS shall do, providing specific, granular requirements for core functionalities such as data ingestion and validation, transformation and storage, metadata management, quality control, data rescue, processing, and user interface interactions. These requirements operationalize the high-level functional components outlined in WMO-No. 1131.
- **Non-Functional Requirements**: Defines how well a CDMS shall perform, covering essential attributes like performance, scalability, reliability, maintainability, usability, interoperability, data governance, and comprehensive security measures. These requirements ensure that a CDMS not only fulfills its

functional duties but also operates effectively, securely, and sustainably in various environments.

- **Appendix**: Provides supplementary information, including a glossary of terms, a list of acronyms used throughout the document, references to related standards and specifications, and any other relevant supporting material that aids in the understanding and application of these specifications.

Each requirement within the functional and non-functional sections is designed to be clear, measurable, and directly applicable, serving as a valuable reference for both system development and compliance assessment activities in the context of WMO-No. 1131.

[1] https://community.wmo.int/governance/commission-membership/commission-observation-infrastructures-and-information-systems-infcom/commission-infrastructure-officers/infcom-management-group/standing-committee-information-management-and-technology-sc-imt

[2] https://community.wmo.int/governance/commission-membership/infcom

Chapter 2. Conformance

This specification defines the following conformance classes corresponding to the functional capabilities of a Climate Data Management System:

- **Core Data Management** (Required): Foundational observation storage and retrieval, provenance metadata management, and basic station and sensor metadata storage. This class provides the data management foundation on which all other conformance classes depend.
- **Quality Control** (Required): Climate-specific QC procedures including range, temporal, spatial, and climatological checks
- **Data Ingestion** (Required): Multi-format data import, validation, and normalization
- **Data API** (Required): Standards-based data query, retrieval, and distribution interfaces
- **Homogenization** (Recommended): Statistical detection and adjustment of inhomogeneities in climate time series
- **Data Rescue** (Optional): Digitization workflow for historical paper records including imaging and data entry
- **Metadata Management** (Required): Advanced station and element metadata management including WIGOS compliance, complete station history, discovery metadata, and metadata quality validation
- **User Interface** (Required): Interactive visualization, data review, and quality control validation capabilities
- **Climate Products** (Recommended): Production and publication of standard climate products and indices
- **Processing Automation** (Recommended): Scheduled and automated execution of routine data processing workflows

NOTE

The classification of conformance classes as Requirement, Recommendation and Permission aligns with the Required, Recommended and Optional classifications respectively from WMO-No. 1131. Required classes must be implemented for a system to be considered conformant. Recommended classes should be implemented where operationally feasible. Optional classes may be implemented based on specific organizational needs.

Chapter 3. References

The following documents are referred to in this specification:

<To follow>

Chapter 4. Terms and definitions

For the purposes of this document, the terms and definitions given in WMO-No. 1131 and the following apply:

<To Follow>

Chapter 5. Conventions

For the purposes of this document, the conventions described in this clause apply.

5.1. Identifiers

The normative provisions in this module are denoted by URIs which follow a hierarchical structure to ensure unique identification and traceability.

The base URI for this specification is: `spec-name/[version]/[module-name]`

All requirements that appear in this document are denoted by **partial URIs** which are relative to this base.

The structure for partial URIs is defined as follows:

5.1.1. Requirements URIs

Requirements are identified by a URI fragment that includes the Requirements Class name and a unique identifier for the specific requirement.

- **Format:** `/req/[requirements-class-name]/[requirement-id]`
- **Example:** `/req/quality-control/range-check`

5.1.2. Recommendation URIs

Recommendations are identified by a URI fragment that includes the Recommendations Class name and a unique identifier for the specific requirement.

- **Format:** `/rec/[recommendation-class-name]/[recommendation-id]`
- **Example:** `/rec/homogenisation/detection`

> DB comment: need to check the syntax.

5.2. Use of IETF RFC 2119 Keywords

The keywords "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in IETF RFC 2119.

The keywords indicate the level of strictness required for compliance with a normative provision:

- **MUST/REQUIRED/SHALL:** These keywords mean that the definition is an **absolute requirement** of the specification.
- **MUST NOT/SHALL NOT:** These phrases mean that the definition is an **absolute prohibition** of the specification.
- **SHOULD/RECOMMENDED:** These keywords mean that there may exist valid reasons in particular circumstances to ignore a particular item, but the full implications must be understood and carefully weighed before choosing a different course.
- **SHOULD NOT/NOT RECOMMENDED:** These phrases mean that there may exist valid reasons in

particular circumstances when the behavior is acceptable or even useful, but the full implications should be understood and the case carefully weighed before proceeding.

- **MAY/OPTIONAL:** These keywords mean that an item is truly optional.

5.3. Typographical and Notational Conventions

The following typographical conventions are used throughout this document to distinguish different types of content:

- `Monospace Text`: Used for source code examples, file names, directory names, machine-readable data structures (e.g., JSON keys, XML tags), URIs, and literal text that appears in a program or command line.
- ***Italics***: Used for titles of documents, variable placeholders within code or URI templates, and for emphasis.
- **[Bracketed] Text**: Used for placeholders in this template (e.g., `[spec-name]`, `[module-name]`) that must be replaced with specific content during implementation.
- **[NOTE], [TIP], [IMPORTANT]**: Used to highlight informative text blocks that provide context, advice, or warnings that are **not** part of the normative requirements.

Chapter 6. Functional Requirements

6.1. Requirements Class: Core

Requirements class	/req/core
Target type	Software system
Dependency	None

6.1.1. Overview

The Core requirements class defines the requirements for basic observation storage, retrieval, and metadata management capabilities. This provides the foundational data management capabilities for a Climate Data Management System and support for long-term data stewardship.

6.1.2. Justification and Regulatory Basis

WMO-No. 1131 identifies Core Data Management as a Required component across all CDMS implementations. WMO-No. 1131 further requires support for observations from across all Earth system domains, necessitating that a CDMS handle data from both fixed-position stations and mobile platforms.

The Manual on High-quality Global Data Management Framework for Climate (WMO-No. 1238) requires Members to "ensure that Entities establish and document data management capabilities and practices for their climate data" (3.1.1). The framework's principles mandate "preservation and management of all data and metadata that are used for, or may potentially be useful for, climate studies" (1.1.1(3)). Core capabilities support the FAIR data principles requiring data to be Findable, Accessible, Interoperable and Reusable (3.2).

6.1.3. Requirements

6.1.3.1. Observation Data Model

Requirement 1	
Identifier	/req/core/data-model
Statement	The system SHALL implement a data model capable of storing in-situ point observations from across the Earth system domains and disciplines. The data model SHALL include: • Station and sensor metadata • Observed element/variable • Observation value/result and associated units • Quality flag status • Observation timestamp, including time zone support • Geographic location of the observation (derived from station metadata for fixed stations; stored per-observation for mobile platforms) • Provenance metadata

NOTE A draft data model <link to wccdm-obs>

NOTE	The definition of an observed element requires careful consideration of all dimensions that distinguish one measurement from another, including: measurement type (e.g. temperature, pressure), sensor height or depth, aggregation duration, statistical function (instantaneous, mean, maximum, minimum), and unit of measure. Systems should ensure that element definitions are sufficiently granular to avoid accidental overwriting of observations from different sensors or measurement configurations. Source data provenance (e.g. transmission format or network) should be recorded separately rather than embedded in the element definition.

NOTE	Support for observations from across the Earth system domains is a core requirement of this specification. This includes both fixed-position stations and mobile platforms (ships, buoys, radiosondes, aircraft). For fixed stations, geographic location is derived from station metadata. For mobile platforms, geographic location must be stored with each individual observation.

6.1.3.2. Reference data

- To follow

6.1.3.3. Observation Storage

Requirement 2	
Identifier	/req/core/observation-storage
Statement	The system SHALL provide persistent storage for observation records.

6.1.3.4. Observation Retrieval

Requirement 3	
Identifier	/req/core/observation-retrieval
Statement	The system SHALL support retrieval of observations filtered by: • Station identifier or group of stations • Geographic bounding box or polygon • Time period (start and end timestamps) • Observed element/variable • Quality control status

6.1.3.5. Station and Sensor Metadata Storage

Requirement 4	
Identifier	/req/core/metadata-storage
Statement	
A	The system SHALL provide persistent storage for station and sensor metadata as specified in /req/metadata-management/wigos.

B	The system SHALL support extensibility mechanisms to store metadata fields not covered by the WIGOS Metadata Standard, including national identifiers, legacy identifiers from predecessor networks, and other organisation-specific metadata attributes.

6.1.3.6. Station and Sensor Metadata Retrieval

Requirement 5	
Identifier	/req/core/metadata-retrieval
Statement	The system SHALL support retrieval of station and sensor metadata filtered by: • Station identifier (WIGOS Station Identifier, national identifier, or legacy identifier) • Geographic extent (bounding box, polygon, or proximity search) • Administrative boundaries (country, region, organization) • Station classification or network affiliation • Observed element capability • Operational status (active, historical, planned) • Temporal validity (metadata state at a specified point in time)

6.1.3.7. Station and Sensor Metadata Versioning

Requirement 6	
Identifier	/req/core/metadata-versioning
Statement	The system SHALL maintain temporal versioning of all station and sensor metadata such that: • All changes to metadata are recorded with effective date and time • Historical states of metadata can be reconstructed for any point in time • The system can return metadata valid at the time of any given observation • Metadata changes do not overwrite or destroy previous versions

6.1.3.8. Provenance Metadata Storage

Requirement 7	
Identifier	/req/core/provenance-storage

Statement	The system SHALL store provenance metadata for all observation data documenting the complete data lineage including: • Original data source identification (network, station, transmission method) • Acquisition timestamp (when data entered the system) • Ingestion processing details (format conversion, unit normalization, coordinate transformation) • Quality control processing history (quality control procedures applied, flags assigned, timestamps) • Manual interventions (edits, corrections, deletions with operator identification and justification) • Derived data lineage (source observations, algorithms, parameters used in derivation) • Homogenisation adjustments (break points detected, adjustment factors applied, reference series used) • Data rescue provenance (source document reference, image identifier, keying method, verification status)

6.1.3.9. Provenance Metadata Retrieval

Requirement 8	
Identifier	/req/core/provenance-retrieval
Statement	The system SHALL support retrieval of provenance metadata including: • Complete processing history for any observation or set of observations • Identification of all transformations applied to data from acquisition to current state • Audit trail queries by operator, time period, or processing type • Traceability from derived products back to original source observations • Export of provenance information in machine-readable format (e.g., W3C PROV-O, ISO 19115 lineage)

6.1.3.10. Provenance Metadata Immutability

Requirement 9	
Identifier	/req/core/provenance-immutability
Statement	The system SHALL ensure provenance records are immutable such that: • Provenance entries cannot be modified or deleted after creation • Original (raw) observation values are preserved even when corrections are applied • A complete audit trail exists for all data modifications

6.1.3.11. Batch-processing

Requirement 10	
Identifier	/req/core/batch-processing
Statement	The system SHALL support batch operations for efficiency: • Bulk station metadata updates via file upload • Batch quality control flag modifications with justification • Bulk data export for multiple stations and time ranges • Progress indicators for long-running operations • Ability to cancel or pause long-running batch operations

6.2. Requirements Class: Observations Quality Control

Requirements class	/req/quality-control
Target type	Software system
Dependency	/req/core

6.2.1. Overview

The Observations Quality Control requirements class defines requirements to ensure observation data meets quality standards for long-term environmental records through systematic automated checks that detect erroneous, suspicious, or inconsistent values while maintaining complete audit trails of all quality control decisions. Quality control is essential for establishing trustworthy observational records suitable for long-term trend analysis and monitoring.

6.2.2. Justification and Regulatory Basis

WMO-No. 1131 classifies Observations Quality Control as Required (Section 5.3), emphasizing that "all data have been checked for accuracy, and that a record is maintained of what checks were performed, the outcome of the checks and when the checks took place."

WMO-No. 1238 mandates that "Entities apply quality control to all climate datasets under their management and record the result of that quality control in the metadata" (3.3.2.1).

NOTE	The *Guidelines on Surface Station Data Quality Control and Quality Assurance for Climate Applications* (WMO-No. 1269) provide specific methodologies referenced in both documents.

6.2.3. Requirements

6.2.3.1. Quality Flag Management

Requirement 11	
Identifier	/req/quality-control/flags

Statement	The system SHALL maintain quality flags and: • Support multiple quality control levels (automatic, manual, final) • Preserve quality control history and audit trail • Allow manual override with justification • Record which specific quality control check(s) triggered each flag and the reason for failure

NOTE	Implementers are encouraged to adopt a published flagging convention to facilitate data exchange and interoperability. WMO-No. 1269 provides guidance on quality flag definitions applicable to long-term observational data.

6.2.3.2. Quality Control Configuration Management

Requirement 12	
Identifier	/req/quality-control/configuration
Statement	The system SHALL support configuration of quality control parameters, including: • Station-specific and element-specific thresholds • Seasonal variation of thresholds where climatologically appropriate • Enabling/disabling of individual checks per station or element • Version control of quality control rule configurations with effective dates

NOTE	Thresholds may be either fixed values (e.g. absolute physical limits) or dynamically derived (e.g. based on climatological statistics such as percentiles). Implementations should ensure that simple fixed-threshold configurations remain straightforward to set up, with advanced options such as seasonally varying or statistically derived thresholds available but not required for basic operation.

6.2.3.3. Quality Control Execution

Requirement 13	
Identifier	/req/quality-control/execution
Statement	The system SHALL support triggering of quality control processing: • Automatically upon data ingestion • On demand by an authorized user, including reprocessing of previously checked observations • As a scheduled or batch process, including reprocessing of previously checked observations

6.2.3.4. Range Checks

Requirement 14	
Identifier	/req/quality-control/range-checks

Statement	The system SHALL perform range validation checks, including: • Gross error limits (physically impossible values) • Climatological based range checks • Sensor-based range checks (measurements within observable range for sensor) • Configurable thresholds per element and station

6.2.3.5. Temporal Consistency Checks

Requirement 15	
Identifier	/req/quality-control/temporal-checks
Statement	The system SHALL perform temporal consistency checks, including: • Flatline checks - detection of repeated identical values • Rate of change checks - validation of temporal gradients • Spike checks - detection of anomalous rapid changes followed by rapid reversals

6.2.3.6. Spatial Consistency Checks

Requirement 16	
Identifier	/req/quality-control/spatial-checks
Statement	The system SHALL perform spatial consistency checks including: • Comparison with neighbouring observations using configurable selection criteria, including maximum distance, elevation difference, station type, covariability between stations, and predefined station lists • Spatial regression tests

6.2.3.7. Internal Consistency Checks

Requirement 17	
Identifier	/req/quality-control/internal-consistency
Statement	The system SHALL perform internal consistency checks to validate physical relationships between simultaneously observed elements as specified in WMO-No. 1269.

NOTE	Examples of internal consistency checks include validation of temperature relationships (T_{min} ¬ T_{avg} ¬ T_{max}), humidity constraints, pressure consistency, and precipitation-weather code agreement. WMO-No. 1269 provides comprehensive guidance on element-specific checks.

NOTE	Some internal consistency checks may also be applied within Data Collection and Processing Systems (DCPS) during real-time processing. The CDMS applies these checks in the context of quality assurance for long-term observational records, where access to full station metadata and the complete observation history allows for more thorough validation than is typically possible in real-time operational systems.

6.3. Requirements Class: Data Ingestion

Requirements class	/req/data-ingestion
Target type	Software system
Dependency	/req/core

6.3.1. Overview

The Data Ingestion requirements class defines requirements for the reliable ingestion of observational data from diverse sources and formats. This requirements class is restricted to observational (in-situ) data from stations and does not cover the ingestion of gridded or raster data products (e.g., satellite imagery, reanalysis fields, radar composites). Data ingestion serves as the entry point for observations into the observational record, requiring careful handling to preserve data integrity through validation and normalization while maintaining complete provenance of source data characteristics.

6.3.2. Justification and Regulatory Basis

WMO-No. 1131 identifies Data Ingest as a Required capability (Section 5.1.1), stating systems must handle "WMO messages, vector, raster array, ASCII and other formats" with "business rules, status logging, automated self-recovery and transformation." This specification restricts the scope of data ingestion to observational (in-situ) data formats; raster and gridded data ingestion is outside the scope of this specification.

WMO-No. 1238 requires Members ensure "Entities encode data, and associated metadata, intended for international exchange following the standards specified in the Manual on Codes (WMO-No. 306)" (3.2.2.1).

6.3.3. Requirements

6.3.3.1. Automated Data Collection

Requirement 18	
Identifier	/req/data-ingestion/automated-collection
Statement	
A	The system SHALL support the following data collection triggers: • **Scheduled polling**: Configurable polling of external data sources at defined intervals • **Real-time subscription**: Subscription to message brokers (e.g., MQTT) for event-driven data collection • **File system monitoring**: Automatic detection and ingestion of new files in monitored directories
B	The system SHALL support the following data transfer protocols: • FTP and SFTP • HTTP/HTTPS • MQTT (for WIS 2.0 message broker integration) • RESTful API calls to external observation systems with authentication management

C	For each automated collection source, the system SHALL maintain: • Collection schedule or subscription configuration • Connection credentials and authentication parameters (stored securely) • Last successful collection timestamp • Collection attempt history and status • Error logs for failed collection attempts
D	The system SHALL implement resilient collection mechanisms: • Retry logic with configurable retry count and backoff intervals • Detection of duplicate data from multiple collection attempts • Graceful handling of temporarily unavailable sources • Alert generation on persistent collection failures • Automatic recovery and resumption after system restart

NOTE	Automated collection sources should be configurable without system restart. The system should support pausing and resuming collection from specific sources for maintenance purposes.

6.3.3.2. Collection Monitoring

Requirement 19	
Identifier	/req/data-ingestion/collection-monitoring
Statement	The system SHALL provide monitoring capabilities for automated data collection: • Dashboard showing status of all configured collection sources • Statistics on collection success/failure rates per source • Data volume metrics (records/observations collected per time period) • Latency metrics (time between observation and ingestion) • Alert generation when expected data does not arrive within defined time windows • Collection log export for auditing and troubleshooting

6.3.3.3. Manual Data Acquisition

Requirement 20	
Identifier	/req/data-ingestion/manual-acquisition

Statement	The system SHALL support manual data submission including: • Web-based file upload interface with drag-and-drop support • Batch upload of multiple files • Upload of compressed archives (ZIP, TAR.GZ) • Manual submission via API with authentication • Upload progress indication for large files • Validation feedback during or immediately after upload • Confirmation of successful ingestion with record counts

6.3.3.4. Format Support

Requirement 21	
Identifier	/req/data-ingestion/formats
Statement	The system SHALL support the ingestion of observational data in the following formats: • **CSV and delimited text files**: With configurable field mapping, delimiter specification (comma, tab, semicolon, pipe), header row handling, and support for common date/time formats • **BUFR**: WMO standard binary format (WMO-No. 306, Volume I.2), supporting ingestion of hourly observations, daily summaries (including DAYCLI report types), and monthly summaries • **WMO text-based formats and code forms**: Including FM-12 SYNOP and FM-71 CLIMAT, and other code forms as specified in WMO Manual on Codes (WMO-No. 306, Volume I.1)

NOTE	Implementations should document which BUFR report types and WMO code forms are supported. The Manual on Codes (WMO-No. 306) defines the complete set of applicable report types and code forms.

Recommendation 1	
Identifier	/rec/data-ingestion/formats
Statement	The system SHOULD support the following additional formats: • **NetCDF** following CF (Climate and Forecast) conventions for station (point) data using discrete sampling geometries • **JSON** following a defined schema for observational data exchange • **XML** following a defined schema or profile used in national or regional networks

NOTE	For JSON, XML, and NetCDF formats, implementations should adopt or reference a specific schema or profile rather than providing generic format support, as the absence of a defined structure makes generic readers impractical. Support for additional historical or legacy formats may be required based on data rescue activities and national archive requirements; these should be documented in conformance documentation.

6.3.3.5. Format Validation

Requirement 22	
Identifier	/req/data-ingestion/validation
Statement	
A	The system SHALL validate ingested data for: • Format compliance • Required field presence (station ID, timestamp, observed element, value) • Data type correctness (numeric values, valid date/time formats) • Referential integrity (station IDs exist in metadata repository) • Value range plausibility (preliminary sanity checks before full QC)
B	Validation failures SHALL result in: • Rejection at the finest possible granularity: individual elements where the format supports it, individual records where element-level rejection is not possible, or the entire file only where record-level rejection is not feasible • Generation of detailed error reports identifying specific validation failures • Logging of validation errors for system monitoring and diagnostics

NOTE The value range plausibility check in Statement A is a lightweight gate applied at ingestion time to catch obviously erroneous values (e.g., temperature of 9999°C) before data enters storage. It is not a substitute for the full quality control checks defined in /req/quality-control, which are applied after ingestion using complete station metadata, climatological context, and inter-element relationships. Data that passes ingestion validation but fails full QC will be stored with appropriate quality flags rather than rejected at the ingestion boundary.

6.3.3.6. Data Normalisation

Requirement 23	
Identifier	/req/data-ingestion/normalization
Statement	
A	The system SHALL normalize ingested data from all supported formats into a common internal representation that includes: • Observation value with standardized unit of measure • Observation timestamp normalized to UTC • Station identifier (preferentially WIGOS Station Identifier) • Observed element code using WMO BUFR parameter codes or an equivalent controlled vocabulary • Source format identifier • Ingestion timestamp and processing metadata

B	The system SHALL perform unit conversions where necessary, maintaining: • Original value and unit in provenance metadata • Conversion algorithm and factors used • Target standard units as specified in WIGOS metadata standards

NOTE

The normalization approach described here — converting all ingested data to a common internal representation — is one valid architectural pattern. Implementations that store data in source formats with format-specific adapters for API access are also conformant, provided that equivalent provenance and retrieval capabilities are met.

Note that WMO BUFR parameter codes do not cover all observable elements. Implementations should document the controlled vocabulary used for element codes and provide mappings for any elements not covered by the BUFR code tables.

6.3.3.7. Duplicate Detection

Requirement 24	
Identifier	/req/data-ingestion/duplicate-detection
Statement	
A	The system SHALL detect and handle duplicate observations based on: • Station identifier • Sensor identifier (where available) • Observation timestamp • Observed element
B	When duplicates are detected, the system SHALL: • Flag duplicate records for review • Apply configurable duplicate resolution strategies (keep first, keep last, manual review) • Log duplicate detection events • Preserve information about all duplicate instances in provenance metadata

NOTE

Operational data exchange systems (including GTS and WIS 2.0) frequently retransmit messages containing identical values. Where duplicate records have identical observed values, automatic suppression without manual review is acceptable. Duplicate resolution requiring manual review should be reserved for cases where values differ between instances.

6.3.3.8. Character Encoding

Requirement 25	
Identifier	/req/data-ingestion/character-encoding

A	The system SHALL handle international character sets: • UTF-8 as default character encoding for all text data • Support for legacy encodings (ASCII, ISO-8859-1) during data ingestion • Proper handling of diacritics and non-Latin scripts in station names and metadata • Character encoding declaration in all text-based exports

6.3.3.9. Calculation of derived parameters

NOTE	Requirements for the calculation of derived parameters are addressed in the Climate Products requirements class: aggregated summary statistics (daily minima/maxima, monthly totals, period means) and climate normals in /req/climate-products/statistics and /req/climate-products/normals; indices such as SPI and CLIMDEX in /rec/climate-products/indices. Scheduling and automation of these computations is addressed in /req/automation.

6.4. Requirements Class: Data API

Requirements class	/req/data-api
Target type	Software system
Dependency	/req/core, /req/security

6.4.1. Overview

Provide standardized programmatic access to climate observation data and metadata, enabling interoperability with external systems and supporting diverse user communities through well-documented APIs. Standard APIs facilitate integration with WIS 2.0, Regional Climate Centres, and international archives.

6.4.2. Justification and Regulatory Basis

WMO-No. 1131 classifies Climate Data Delivery Services as Required (Chapter 8), emphasizing "open spatial standards" and "WMO formats" for data exchange.

WMO-No. 1238 requires that "data, and associated metadata, are registered in the WMO Information System Global Discovery Catalogue" (3.2.1.1c) and that systems support "API interoperability" using standards like OGC API family (Features, Records, Environmental Data Retreival, Processes). The framework principle that "datasets designated to be exchanged internationally through the WMO Information System" (1.1.1(4)) requires standardized API access.

6.4.3. Requirements

6.4.3.1. RESTful API

Requirement 26	
Identifier	/req/data-api/rest

Statement	The system SHALL provide a RESTful API that: • Uses HTTP/HTTPS protocols • Implements standard HTTP methods (GET, POST, PUT, DELETE) where appropriate • Returns data in standard formats (JSON, XML, CSV, NetCDF) • Provides comprehensive API documentation • Supports content negotiation based on Accept headers

NOTE	Implementations should document the intended access scope of the API (e.g., internal/LAN access only, or publicly accessible). Publicly accessible APIs should apply appropriate authentication and rate limiting in accordance with /req/data-api/auth.

NOTE	Implementations are encouraged to provide API documentation in OpenAPI (formerly Swagger) format, which enables interactive documentation and client code generation. OpenAPI specifications should be made available at a well-known endpoint (e.g. /api or /openapi.json).

6.4.3.2. Standards Compliance

Recommendation 2	
Identifier	/rec/data-api/standards
A	The system SHOULD implement one or more of the following standard APIs: • OGC API - Environmental Data Retrieval (EDR) • OGC API - Features • OGC SensorThings API

NOTE	Implementation of standardized APIs enhances interoperability with the broader climate and meteorological data ecosystem. The choice of which standards to implement should be based on user community needs and integration requirements.

6.4.3.3. Query Capabilities

Requirement 27	
Identifier	/req/data-api/query
Statement	The system SHALL support API queries with the following parameters: • Spatial extent (bounding box, point, polygon) • Temporal extent (time range, specific timestamp) • Station identifiers (single or multiple) • Observed elements/variables • Quality control status • Aggregation level (raw observations, or pre-computed official statistics where available)

NOTE	On-demand statistical aggregation by the API (e.g., computing daily means from raw observations at query time) is not recommended. Rigorous statistics require careful handling of data gaps, quality flags, and defined computation periods. Pre-computed official statistics should be stored as standard derived observations and served through the same query interface.

6.4.3.4. Authentication and Authorization

Requirement 28	
Identifier	/req/data-api/auth
Statement	Authentication and authorization SHALL comply with /req/security/authentication and /req/security/authorization

6.4.3.5. API Versioning

Requirement 29	
Identifier	/req/data-api/versioning
Statement	The system SHALL implement API versioning to: • Maintain backward compatibility with existing clients • Clearly indicate API version in endpoints or headers • Provide migration guidance for deprecated API versions • Support multiple API versions concurrently during transition periods

6.4.3.6. Integration Interfaces

Requirement 30	
Identifier	/req/data-api/integration
Statement	The system SHALL provide integration mechanisms for: • WIS 2.0 message broker subscription and publication • Webhook notifications for data availability or quality control events • File-based integration using watched directories or object storage • Message queue integration for asynchronous processing

6.4.3.7. Data Format Interoperability

Requirement 31	
Identifier	/req/data-api/data-formats
Statement	

A	The system SHALL export data in interoperable formats: • NetCDF following CF (Climate and Forecast) conventions for station (point) data using discrete sampling geometries • CSV with RFC 4180 compliance and documented column definitions • JSON following consistent schema conventions • GeoJSON for observations with spatial context • BUFR for exchange with WMO systems, including CLIMAT and DAYCLI message types for monthly and daily summary data
B	Each export format SHALL be accompanied by documentation including: • Description of the data structure and schema • Units of measure for all variables • Quality flag definitions and values • Temporal reference system (UTC or clearly specified) • Spatial reference system (coordinate system, datum)

NOTE	For high-frequency or real-time data feeds, including full format documentation with every export response may be impractical. In such cases, it is sufficient to make the format definition and schema documentation available via a persistent, well-known endpoint referenced in the API documentation.

6.5. Requirements Class: Homogenisation

Requirements class	/req/homogenisation
Target type	Software system
Dependency	/req/core, /req/quality-control

6.5.1. Overview

Detect and correct non-climatic discontinuities in long-term observation series caused by changes in instrumentation, observation practices, station location, or surrounding environment, ensuring temporal consistency essential for long-term trend analysis. Homogenisation is critical for producing reliable assessments of environmental variability and change.

6.5.2. Justification and Regulatory Basis

WMO-No. 1131 classifies Data Homogenisation as Recommended (Section 6.1.3), recognizing that homogenised datasets "aim to ensure that the only variability remaining in the time series is that resulting from actual climate variability" (4.5.1.3).

WMO-No. 1238 requires that "Entities record any homogenisation or reprocessing carried out on the data as part of the provenance metadata" (3.3.2.5). The Guidelines on Homogenisation (WMO-No. 1245) provide detailed methodologies. GCOS Monitoring Principle 1 requires assessing "impact of new systems or changes to existing systems", making homogenisation essential for maintaining long-term climate records.

6.5.3. Requirements

NOTE

These requirements apply to all observable elements where sufficient station density and record length exist to support homogenisation. While temperature is the most commonly homogenised element, the requirements apply equally to precipitation, pressure, wind, and other elements where applicable.

Homogenisation methods are most reliably applied to monthly data. Application to daily or sub-daily data is possible but introduces greater uncertainty; implementations should document the temporal resolution to which homogenisation has been applied.

Implementations may satisfy these requirements either by integrating homogenisation algorithms directly within the CDMS, or by providing well-defined interfaces for data exchange with specialist third-party homogenisation tools, provided that the documentation and data management requirements of this class are met.

6.5.3.1. Break Point Detection

Recommendation 3	
Identifier	/rec/homogenisation/detection
Statement	The system SHOULD: • Implement statistical methods for detecting inhomogeneities. • Support: - Relative homogenisation using reference series from neighboring stations - Absolute homogenisation methods where appropriate - Detection of multiple break points in a single series - Configurable significance levels for break point detection

6.5.3.2. Adjustment Calculation

Recommendation 4	
Identifier	/rec/homogenisation/adjustment
Statement	The system SHOULD calculate adjustments for detected inhomogeneities: • Compute adjustment factors based on the magnitude of detected breaks • Apply adjustments that preserve long-term means and variability characteristics • Support different adjustment methods (additive, multiplicative, or hybrid) • Validate adjusted series for statistical consistency

6.5.3.3. Homogenisation Documentation

Requirement 32	
Identifier	/req/homogenisation/documentation

Statement	The system SHALL document all homogenisation processes applied including: • Break points detected with dates and statistical confidence levels • Adjustment factors applied to each segment • Reference series used for relative homogenisation • Statistical test results and significance values • Comparison of original and homogenised series • Metadata about known station changes that correspond to detected breaks

6.5.3.4. Homogenised Data Management

Requirement 33	
Identifier	/req/homogenisation/data-management
Statement	The system SHALL manage homogenised data such that: • Original (raw) observations are preserved unchanged • Homogenised values are stored as derived data with clear lineage • Both original and homogenised series are accessible to users • Provenance metadata clearly identifies homogenisation processing • Homogenisation can be revised or reprocessed without data loss

6.6. Requirements Class: Data Rescue

Requirements class	/req/data-rescue
Target type	Software system
Dependency	/req/core, /req/data-ingestion

6.6.1. Overview

Facilitate the recovery of historical observations from non-digital sources, ensuring rescued data is integrated into the observational records with complete provenance and appropriate quality control. Data rescue extends the temporal coverage of observational records, essential for understanding long-term variability and change.

6.6.2. Justification and Regulatory Basis

WMO-No. 1131 classifies Data Rescue capabilities as Recommended (Section 5.2), covering "imaging and digitization, monitoring, and data entry."

WMO-No. 1238 requires Members ensure "Entities develop a data rescue strategy" including "systematic" preservation activities, digitization plans, and resource mobilization (3.5.3.1).

NOTE	The Guidelines on Good Practices for Data Rescue (WMO-No. 1182) emphasizes that data rescue is "the ongoing process of identifying and preserving all data, records and climate archives that are at risk of being lost."

6.6.3. Requirements

6.6.3.1. Document Imaging Management

Recommendation 5	
Identifier	/rec/data-rescue/imaging
Statement	The system SHOULD support document imaging workflows including: • Storage of scanned images with persistent identifiers • Association of images with source document metadata (archive, collection, date range) • Support for multiple image formats (JPEG, TIFF, PDF) • Image quality metadata (resolution, color depth, scan date) • Linkage between images and digitized observations

NOTE	Automated Optical Character Recognition (OCR) of historical documents is outside the scope of this specification. The data entry workflows described here are based on manual keying by operators, with source images displayed alongside data entry forms to support transcription. OCR tools may be used as a supplementary aid at the implementer's discretion.

6.6.3.2. Data Entry Interface

Recommendation 6	
Identifier	/rec/data-rescue/data-entry
A	The system SHOULD provide a data entry interface for manual keying that: • Displays source images alongside data entry forms • Supports configurable data entry templates based on historical form layouts • Provides real-time validation of entered values • Tracks data entry operator and timestamp • Supports both single-key and double-key entry workflows for quality assurance

6.6.3.3. Data Rescue Quality Control

Requirement 34	
Identifier	/req/data-rescue/quality-control
Statement	The system SHALL apply quality control to rescued data including: • Comparison between double-key entries to identify transcription errors • Range and consistency checks appropriate to historical observation practices • Verification against original images when discrepancies are detected • Flagging of uncertain or illegible values • Documentation of data rescue quality assurance decisions

NOTE	Once digitized observations have passed data rescue quality assurance, they SHALL be submitted for ingestion through the same processing pipeline as contemporary observations, including the full suite of quality control checks defined in /req/quality-control.

6.6.3.4. Data Rescue Provenance

Requirement 35	
Identifier	/req/data-rescue/provenance
Statement	The system SHALL record data rescue provenance including: • Source document identification and archive location • Image identifier and scan metadata • Data entry operator(s) and keying date(s) • Keying method (single-key, double-key, automated OCR) • Quality assurance verification status • Known limitations or uncertainties in rescued data

6.7. Requirements Class: Metadata Management

Requirements class	/req/metadata-management
Target type	Software system
Dependency	/req/core, /req/security

6.7.1. Overview

Maintain comprehensive metadata supporting data interpretation, quality control, homogenisation, and fitness-for-purpose assessment. This encompasses station and sensor metadata (WIGOS), formal definitions of observable elements, dataset-level discovery and lineage metadata, dataset quality reporting, access constraints, and acquisition provenance. Comprehensive metadata is essential for establishing the context and trustworthiness of observational records.

6.7.2. Justification and Regulatory Basis

WMO-No. 1131 classifies Climate Metadata management as Required (Section 4.3 and 5.5), encompassing "observations metadata, discovery metadata, and data provenance metadata."

WMO-No. 1238 requires Entities to create and maintain three types of metadata: discovery metadata (per WMO-No. 1060, Volume II), observations metadata (per WMO No. 49, Volume I), and provenance metadata, defined as "information documenting the origin, processing history, and all modifications to a dataset throughout its lifecycle."

For findability and discovery (3.2.1.1), Entities SHALL assign globally unique and persistent identifiers to data and metadata (a), describe data with WIS-conformant metadata (b), and ensure data and metadata are registered in the WMO Information System Global Discovery Catalogue (c). For reusability (3.2.3), Entities SHALL include a usage licence with recommended data published through WIS (3.2.3.2), and any licence for core data SHALL be compatible with the WMO Unified Data Policy (3.2.3.3).

Section 3.3.1 establishes provenance requirements: Entities SHOULD maintain comprehensive records of

all modifications and processing throughout the data lifecycle (3.3.1.1), including the origin of observations, processing history, and methodology references (3.3.1.2). Entities SHALL collect observational metadata and include it in provenance records (3.3.1.4), with reference to the WIGOS Metadata Standard (WMO-No. 1192) and the Manual on the WMO Integrated Global Observing System (WMO-No. 1160). Entities SHALL include a full version history for critical datasets used to monitor climate change (3.3.1.6). Section 2.1.2.3 further requires that Entities responsible for making observations follow the provisions for collection, management, and publication of observational metadata as specified in WMO-No. 1160.

Section 3.3.2 establishes quality assurance metadata requirements: Entities SHALL apply quality control to all datasets and record results in metadata (3.3.2.1), SHALL assign quality flags or indicators to all records (3.3.2.2), SHOULD update metadata with assessed quality at each stage of the quality control process (3.3.2.3), and SHALL record any homogenisation or reprocessing in provenance metadata with links to processing documentation (3.3.2.5).

The ISO 19115 family of standards (ISO 19115-1, ISO 19115-2, ISO 19157, ISO 19110) provides widely recognised concepts for geographic dataset metadata, including lineage, data quality, constraints, maintenance, and feature cataloguing. These concepts inform the requirements in this class; however, as ISO standards are not freely available, they are referenced informatively rather than as normative dependencies. Freely available standards are used as normative references throughout, including the WMO WIGOS Metadata Standard (WMO-No. 1192), WCMP2, W3C PROV-O (provenance), and the CF Conventions for Climate and Forecast Metadata.

6.7.3. Requirements

6.7.3.1. WIGOS Metadata Compliance

Requirement 36	
Identifier	/req/metadata-management/wigos
Statement	
A	The system SHALL support metadata fields as specified in WIGOS Metadata Standard (WMO-No. 1192), including: • Station identification (WIGOS Station Identifier, national identifiers, historical identifiers) • Station name, description, and alternative names • Geographic location (latitude, longitude, elevation, datum reference) • Administrative information (operating organization, country, region) • Station classification (WMO station type, network affiliations, programme participation) • Sensor and instrument inventory with make, model, serial number, and calibration status • Measurement configuration (observed elements, measurement height/depth, sampling interval) • Siting environment description and exposure classification • Operating schedule and reporting practices • Responsible party and contact information

B	The system SHALL support extensibility mechanisms to store metadata fields not covered by the WIGOS Metadata Standard, including national identifiers, legacy identifiers from predecessor networks, and other organisation-specific metadata attributes.

6.7.3.2. Station History Documentation

Requirement 37	
Identifier	/req/metadata-management/history
Statement	The system SHALL:
A	maintain a complete station history, including: • Station relocations with old and new coordinates • Instrument changes with make, model, and installation dates • Changes to observation practices or measurement heights • Changes to surrounding environment affecting exposure • Periods of station closure or data gaps • Network affiliation changes
B	for each entry in the station history include: • Effective date and time of the change • Description of the change • Reason for the change where known • Responsible party documenting the change • Supporting documentation references

NOTE	The station history (Statement A) combined with the station and sensor metadata maintained under /req/metadata-management/wigos provides full dataset-to-instrument traceability: for any observation timestamp, the active instrument configuration can be determined from the instrument change records and their effective dates. This information supports reproducibility and uncertainty assessment without requiring a separate acquisition metadata record.

6.7.3.3. Element and Feature Catalogue

Requirement 38	
Identifier	/req/metadata-management/feature-catalogue

Statement	The system SHALL maintain a catalogue of observable elements, recording for each element: • A unique element identifier and human-readable name • Measurement class (e.g., temperature, precipitation, pressure, wind speed) • Observation height or depth relative to the surface • Accumulation or averaging duration • Unit of measure • Statistical function (instantaneous, mean, maximum, minimum, accumulated total, etc.) • Applicable value range and resolution • Associated controlled vocabulary reference: a WMO BUFR parameter code (from the freely available WMO code tables) or a CF standard name (from the freely available CF Conventions) where applicable

NOTE

The combination of measurement class, observation height/depth, duration, unit, and statistical function constitutes the unique identity of an observable element. Two observations that differ in any of these properties represent distinct elements and must be stored and described separately. This ensures unambiguous interpretation of stored values across stations and datasets. The element identity concept is also described in the data model NOTE in /req/core/data-model; the feature catalogue requirement here provides the normative basis for maintaining that element dictionary as a managed, queryable catalogue separate from the individual observation records.

The concept of a formal element catalogue is informed by ISO 19110 (Feature Cataloguing) and the Observable Property concept in OGC Observations and Measurements. Those standards are referenced informatively; the normative vocabulary references are WMO BUFR code tables (https://library.wmo.int/records/item/35713-manual-on-codes-volume-i-1) and the CF standard name table (https://cfconventions.org/standard-names.html), both of which are freely available.

6.7.3.4. Discovery Metadata

Requirement 39	
Identifier	/req/metadata-management/discovery

A	The system SHALL maintain dataset-level discovery metadata records compliant with the WMO Core Metadata Profile version 2 (WCMP2). Each record SHALL include all mandatory WCMP2 properties: • A unique record identifier following the URN pattern `urn:wmo:md:{centre-id}:{local-identifier}` • Resource title and free-text description • Spatial extent (bounding box in WGS84) and temporal extent • Earth system discipline classification using the WIS2 Topic Hierarchy vocabulary (`properties.themes`) • Responsible organisation name and contact role (`properties.contacts`) • WMO data policy classification (`core` or `recommended`) in `properties.wmo:dataPolicy` • At least one link to a data access endpoint
B	The system SHALL maintain dataset lineage metadata following the W3C PROV Data Model, including: • A lineage statement summarising the overall processing history of the dataset • Source data references: originating station networks, instruments, and observation periods • A record for each significant processing stage applied to the dataset, covering at minimum: - Ingestion and format normalisation - Quality control procedures and the version or configuration applied - Homogenisation adjustments, where applicable - Statistical aggregation or derived product computation, where applicable • For each processing stage: date, description, responsible party, and software or method reference
C	The system SHALL support submission and update of WCMP2 records to the WIS2 Global Discovery Catalogue (GDC), either directly or via a configured WIS2 Node.

NOTE

WCMP2 is built on OGC API - Records (GeoJSON) and intentionally does not incorporate the ISO 19115/19139 XML structures used in earlier WMO metadata standards. WCMP2 serves the discovery function: making datasets findable in the WIS2 Global Discovery Catalogue. Dataset lineage metadata, structured following the W3C PROV Data Model (https://www.w3.org/TR/prov-dm/), serves a complementary function: documenting the detailed processing provenance of datasets for scientific interpretation and fitness-for-purpose assessment. The PROV Data Model defines the core concepts of Entity (the dataset or observation series), Activity (a processing step), and Agent (the responsible party or software).

Discovery metadata records are dataset-level descriptions and are distinct from the station and sensor metadata described in /req/metadata-management/wigos.

Two levels of provenance/lineage are maintained in a CDMS: (1) observation-level provenance (/req/core/provenance-storage, /req/core/provenance-immutability) records the complete immutable history of each individual observation — ingestion, QC flags, corrections, and data rescue events; (2) dataset-level lineage (this requirement, Statement

B) summarises the processing history of a collection of observations as a whole, supporting discovery and fitness-for-purpose assessment. Dataset-level lineage is a descriptive summary; it is not a substitute for the immutable observation-level provenance record and does not carry the same immutability constraint.

Export formats are addressed in /req/metadata-management/export.

For datasets classified as `recommended` under the WMO data policy, at least one access link with associated licence information is required by WCMP2.

The concepts of lineage statement, process step, and source data used in Statement B above are informed by the ISO 19115 family (ISO 19115-1 LI_Lineage, ISO 19115-2 LE_ProcessStep); however, those standards are not freely available and are not normative dependencies of this specification.

6.7.3.5. Dataset Quality Reporting

Requirement 40	
Identifier	/req/metadata-management/data-quality
A	The system SHALL maintain dataset-level quality metadata including: • Completeness measures: proportion of expected observations present over the dataset period, covering both omission (records absent) and commission (duplicate or spurious records) • Temporal quality: consistency of timestamps and identification of temporal gaps or overlaps • Thematic accuracy: conformance of values to expected ranges and inter-element consistency, drawing on quality control results from /req/quality-control • A conformance result indicating whether the dataset meets defined quality thresholds
B	The system SHALL generate dataset quality reports that: • Summarise completeness by station, element, and time period • Identify temporal gaps and periods of reduced data availability • Report the distribution of quality control flag statuses across the dataset

NOTE

Dataset-level quality reporting is distinct from per-observation quality flags managed by /req/quality-control. Quality flags on individual observations provide the source data for computing dataset-level quality metrics. These aggregated metrics support fitness-for-purpose assessment and inform users about the suitability of a dataset for specific applications.

Dataset quality metrics may be derived automatically from per-observation quality control results, or entered manually by data managers to document known issues or provide additional context. Dataset quality reporting (Statement B) covers the generation and export of reports drawing on both sources.

The structure and terminology used here (completeness, temporal quality, thematic accuracy, conformance result) are informed by ISO 19157 (Data Quality), which is referenced informatively. ISO 19157 is not freely available and is not a normative dependency of this specification.

6.7.3.6. Dataset Maintenance Information

Recommendation 7	
Identifier	/rec/metadata-management/maintenance
Statement	The system SHOULD record dataset maintenance metadata including: • Update frequency (e.g., real-time, daily, monthly, irregular) • Date of the most recent update • Maintenance scope (observations, metadata, quality flags, or derived products) • Responsible party for dataset maintenance • Notes on planned updates or known data gaps

6.7.3.7. Data Access Constraints

Requirement 41	
Identifier	/req/metadata-management/constraints
Statement	The system SHALL record data access and use constraints for each dataset, consistent with WMO data policy requirements as defined in WMO-No. 1238 and the `wmo:dataPolicy` property specified in WCMP2, including: • WMO data policy classification (core or recommended), consistent with /req/metadata-management/discovery • Legal access constraints: any restrictions on who may access the data • Use constraints: conditions on redistribution or derivative use • Security classification where applicable (e.g., national security requirements) • Embargo periods: dates before which data may not be published externally

NOTE

Constraint metadata must be enforced through the access control mechanisms defined in /req/security/authorization. Embargo periods and redistribution restrictions recorded here should be applied consistently across all data access interfaces, including the API (/req/data-api) and any export functions.

The legal and security constraint concepts used here are informed by ISO 19115-1 (MD_LegalConstraints, MD_SecurityConstraints), referenced informatively. The normative basis is WMO-No. 1238 data policy requirements and the WCMP2 `wmo:dataPolicy` property.

6.7.3.8. Metadata Standards Export

Requirement 42	
Identifier	/req/metadata-management/export

Statement	The system SHALL support export of metadata in standard formats including: • WIGOS Metadata Representation (WMDR) for station and sensor metadata • WMO Core Metadata Profile v2 (WCMP2, GeoJSON) for dataset discovery metadata • W3C PROV-O or equivalent structured format for dataset lineage and provenance records • Dataset quality reports in a documented, structured format (e.g., JSON or XML with defined schema) covering completeness, temporal quality, and quality flag distribution • Element catalogue in a documented structured format referencing WMO BUFR parameter codes or CF standard names • Station metadata exchange formats used by regional or global programmes

NOTE	A simplified second version of the WIGOS Metadata Representation (WMDR) is under development. Implementations should monitor progress and plan for compatibility with the updated version when published.

6.7.3.9. Metadata Quality Validation

Requirement 43	
Identifier	/req/metadata-management/validation
Statement	The system SHALL
A	validate metadata for: • Required field completeness per WIGOS standards • Coordinate system validity and range checks • Temporal consistency (effective dates do not overlap) • Referential integrity (stations referenced by observations exist) • Controlled vocabulary compliance for classification fields
B	generate validation reports identifying: • Missing required metadata • Metadata quality issues or inconsistencies • Recommendations for metadata improvement

6.7.3.10. Citation and Attribution

Requirement 44	
Identifier	/req/metadata-management/citation

Statement	The system SHALL support proper data citation by: • Generating unique persistent identifiers (DOIs or equivalents) for datasets • Providing recommended citation text for datasets • Recording data usage and access patterns for impact assessment • Supporting dataset versioning with clear version identifiers • Maintaining links between derived products and source datasets

6.8. Requirements Class: User Interface

Requirements class	/req/user-interface
Target type	Software system
Dependency	/req/core

6.8.1. Overview

Provide interactive capabilities for scientists, data managers, and users to visualize, explore, and analyze observational data and quality control results. The user interface enables efficient human interaction with complex observational datasets and supports quality assurance workflows.

6.8.2. Justification and Regulatory Basis

WMO-No. 1131 identifies the Graphical User Interface for time-series data exploration as Recommended (Section 7.1), supporting "tables and charts, visualization, manage content, integrated search of climate data, and data download."

While WMO-No. 1238 does not mandate specific UI requirements, it emphasizes the need for "climate scientists to review and validate climate observations, identify inhomogeneities" and for "data managers to manage station metadata and data holdings" (Figure 2), which requires appropriate user interfaces. The requirement for "comprehensive records" and "quality control" workflows (3.3.2) implies interactive tools for human oversight.

6.8.3. Requirements

6.8.3.1. Visualization Capabilities

Requirement 45	
Identifier	/req/user-interface/visualization
Statement	The system SHALL provide data visualization capabilities including: • Time series plots for observation values over configurable time ranges • Simultaneous display of original and quality-controlled or homogenised values • Station location maps with filtering and selection capabilities • Ability to export visualizations in common image formats (PNG, SVG, PDF) • Interactive exploration of quality control flags and metadata

6.8.3.2. Tabular Data Display

Requirement 46	
Identifier	/req/user-interface/tabular
Statement	The system SHALL provide tabular data display capabilities including: • Display of observations in tabular format with configurable columns (timestamp, station, element, value, unit, quality flag) • Sorting and filtering by any displayed column • Pagination for large result sets • Simultaneous display of data from multiple stations or elements for comparison • Export of displayed tabular data in CSV or equivalent flat-file format

6.8.3.3. Integrated Search

Requirement 47	
Identifier	/req/user-interface/search
Statement	The system SHALL provide an integrated search interface that: • Supports filtering by station identifier, station name, geographic area, element, time period, and quality control status • Presents search results with sufficient context (station name, location, data coverage) to identify relevant records • Enables navigation from search results to detailed observation views and station metadata • Supports download of search results

6.8.3.4. Data Entry and Forms

Requirement 48	
Identifier	/req/user-interface/data-entry
Statement	The system SHALL provide form-based interfaces supporting: • Manual entry or correction of observation values with real-time validation • Creation and editing of station and sensor metadata records • Review and resolution of quality control flags and alerts • Assignment of quality control decisions with operator identification recorded in the audit trail

NOTE	Data entry forms in the context of this requirements class support operational workflows such as metadata management, manual observation correction, and QC decision recording. Data entry workflows for historical data rescue (keying from scanned documents) are specified separately in /req/data-rescue.

6.9. Requirements Class: Climate Products

Requirements class	/req/climate-products
Target type	Software system
Dependency	/req/core, /req/quality-control

6.9.1. Overview

Support the computation and management of standard derived products from quality-controlled observation series, including summary statistics, climate normals, and climate indices. Derived products must be clearly distinguished from raw observations with full documentation of their derivation and data lineage.

6.9.2. Justification and Regulatory Basis

WMO-No. 1131 classifies Climate Products as Recommended (Section 6.1), covering the computation of summary statistics, climate normals, and indices from quality-controlled observational records.

WMO-No. 1238 requires that Members ensure the production of standard climate products and that these are made available through appropriate services. The computation of WMO Climate Normals is governed by the WMO Guidelines on the Calculation of Climate Normals (WMO-No. 1203), which prescribes standard 30-year averaging periods. Climate normals and indices support national and international reporting obligations, including contributions to the Global Climate Observing System (GCOS).

6.9.3. Requirements

6.9.3.1. Summary Statistics

Requirement 49	
Identifier	/req/climate-products/statistics
A	The system SHALL support computation and storage of standard summary statistics from quality-controlled observations, including: • Daily summary values: minimum, maximum, mean, and accumulated totals for each observed element • Monthly and annual aggregations of daily summary values • Configurable aggregation periods and functions
B	All computed statistics SHALL be stored as derived observations that: • Reference the source observations used in their computation • Record the computation method, the quality flag thresholds applied, and the proportion of valid source data used • Are clearly distinguished from raw observations in all data access interfaces

NOTE	Rigorous statistical aggregation requires careful handling of data gaps, quality flags, and defined computation periods (which may not align with calendar day boundaries). Statistics computed from series with significant data gaps should be flagged accordingly. Implementations should apply documented minimum data completeness thresholds before producing summary values.

6.9.3.2. Climate Normals

Requirement 50	
Identifier	/req/climate-products/normals
Statement	The system SHALL support computation and storage of climate normals in accordance with WMO guidelines (WMO-No. 1203), including: • Standard 30-year averaging periods as defined by WMO • Monthly and annual normals for each observed element • Recording of the reference period for each set of normals • Ability to compute and store normals for multiple reference periods simultaneously

6.9.3.3. Climate Indices

Recommendation 8	
Identifier	/rec/climate-products/indices
Statement	The system SHOULD support computation of standard climate indices, including: • Temperature and precipitation extremes indices as defined by the Expert Team on Climate Change Detection and Indices (ETCCDI) • Standardised Precipitation Index (SPI) and related drought indices • User-configurable index definitions to accommodate national or regional requirements

6.9.3.4. Product Management

Requirement 51	
Identifier	/req/climate-products/management
Statement	The system SHALL manage derived products such that: • Products are versioned, with earlier versions retained and accessible • Recomputation following correction of source observations generates a new version without overwriting prior versions • Product metadata records the source data period, computation date, and quality constraints applied • Products are accessible through the data API defined in /req/data-api

6.10. Requirements Class: Processing Automation and Scheduling

Requirements class	/req/automation
Target type	Software system
Dependency	/req/core

6.10.1. Overview

Support the scheduled and automated execution of routine data processing workflows, enabling timely and consistent processing without continuous manual intervention. Automation covers quality control processing, statistical product computation, data publication, and system maintenance tasks.

6.10.2. Justification and Regulatory Basis

WMO-No. 1131 classifies Processing Automation as Recommended, emphasizing that routine data processing should be performed systematically and consistently without requiring manual intervention for each cycle.

WMO-No. 1238 requires timely generation of climate products and systematic data processing (2.2.1). The framework principle requiring "comprehensive records of all modifications and processing" (3.3.1.1) implies that automated processing must be fully logged and auditable.

6.10.3. Requirements

6.10.3.1. Scheduled Processing

Requirement 52	
Identifier	/req/automation/scheduling
Statement	The system SHALL support scheduling of routine processing tasks, including: • Time-based scheduling (hourly, daily, monthly, annual, or user-defined intervals) • Event-triggered execution (e.g., automatic quality control following data ingestion) • Configurable processing scope per schedule (station, element, and time period) • Ability to enable and disable schedules without modifying their configuration

6.10.3.2. Processing Workflows

Requirement 53	
Identifier	/req/automation/workflows
Statement	The system SHALL support definition of multi-step processing workflows that: • Chain dependent processing steps in a defined sequence (e.g., quality control ¬ summary statistics ¬ publication) • Halt downstream steps and notify operators when an upstream step fails • Allow manual re-execution of failed or incomplete workflow steps • Support parallel execution of independent processing tasks where applicable

6.10.3.3. Processing Monitoring and Alerting

Requirement 54	
Identifier	/req/automation/monitoring

Statement	The system SHALL provide monitoring and alerting for automated processing, including: • Status visibility of running, completed, and failed tasks • Configurable alerts (email or equivalent notification) when tasks fail or exceed expected duration • Summary view of processing status across all scheduled tasks • Ability for operators to cancel or manually re-trigger processing tasks

6.10.3.4. Processing Audit Log

Requirement 55	
Identifier	/req/automation/audit
Statement	The system SHALL maintain an audit log of all automated processing executions, recording: • Task identifier, type, and configuration at the time of execution • Start time, end time, and completion status • Input data scope: stations, elements, and time period processed • Count of records processed, updated, flagged, or rejected • Any errors or warnings generated during processing

Chapter 7. Non-functional Requirements

This section defines the quality attributes and constraints that specify how well the Climate Data Management System performs its functions. These requirements address system characteristics essential for operational deployment, long-term sustainability, and user satisfaction.

7.1. Requirements Class: Performance and Scalability

Requirements class	/req/performance
Target type	Software system
Dependency	/req/core

7.1.1. Overview

Ensure the system can handle operational data volumes, concurrent users, and query loads efficiently while maintaining acceptable response times and supporting future growth. Performance requirements ensure the CDMS remains viable as data volumes increase and user communities expand over decades of operation.

7.1.2. Justification and Regulatory Basis

WMO-No. 1131 emphasizes that CDMSs must handle "big data" characterized by "large volumes of data in a variety of formats that present challenges for effective data management" (Section 4.1). The specification notes systems must manage "observation datasets spanning minimum 100 years of historical data" with scalability for "considerable amounts of derived data" (Section 4.5).

WMO-No. 1238 requires systems support "duplicate repositories for the duration of their retention periods" (3.4.2.2) and handle data from multiple Earth system domains (1.1.2), implying substantial performance requirements. The framework principle of ensuring "adequate uniformity and standardization" (Introduction, para 5e) requires consistent performance across implementations.

7.1.3. Requirements

7.1.3.1. Query Response Performance

Requirement 56	
Identifier	/req/performance/response-time
Statement	The system SHALL: • Define and document response time targets for common query operations, including single-station data retrieval, multi-station spatial queries, and metadata lookups • Monitor query response times against defined targets • Remain responsive under expected concurrent user load without degradation of correctness or data integrity

NOTE	Response time targets should be defined by implementers based on expected user community size, data volumes, and infrastructure. Targets should be documented in system specifications and reviewed as data volumes and user communities grow. Users

should be informed of long-running queries through progress indication rather than silent waits.

7.1.3.2. Data Volume Capacity

Requirement 57	
Identifier	/req/performance/capacity
Statement	The system SHALL: • Support storage and retrieval of observational records spanning multiple decades of historical data across all managed station networks • Accommodate growth in data volume over the operational lifetime of the system without requiring re-architecture of core data storage • Complete ingestion of observations from all configured sources within defined latency targets appropriate to the data delivery schedule of each source

7.1.3.3. Scalability

Recommendation 9	
Identifier	/rec/performance/scalability
Statement	The system SHOULD be designed to support scaling of processing and storage capacity to accommodate increasing data volumes and user loads as national observing networks grow.

7.2. Requirements Class: Reliability and Availability

Requirements class	/req/reliability
Target type	Software system
Dependency	/req/core

7.2.1. Overview

Ensure the system operates dependably with minimal unplanned downtime, maintains data integrity under all conditions, and recovers gracefully from failures. Reliability is essential for climate data, which serves as an irreplaceable scientific resource requiring protection for future generations.

7.2.2. Justification and Regulatory Basis

WMO-No. 1131 states that "the most important stakeholders for climate data are future generations, and the reason to implement and manage CDMSs now is to protect the integrity of climate data for their future use" (Section 2.3). The specification’s core theme is to "ensure that the data are securely and transparently managed" (Section 2.3).

WMO-No. 1238 mandates that "Members shall ensure that Entities develop and maintain a business continuity plan to mitigate risks associated with disruption of operations" (3.4.1.1) and requires "routine backup, and procedures for timely restoration of their databases" (3.4.1.2). The requirement to "retain their original and irreproducible climate records, and associated metadata, permanently" (3.4.1.4) establishes reliability as a foundational requirement.

7.2.3. Requirements

7.2.3.1. System Availability

Requirement 58	
Identifier	/req/reliability/availability
Statement	The system SHALL: • Define and publish availability targets for data retrieval and ingestion services • Monitor system availability and report against defined targets • Communicate scheduled maintenance windows to users in advance • Minimise scheduled maintenance downtime through rolling update or equivalent procedures • Notify administrators promptly of unplanned outages with sufficient detail for diagnosis

NOTE

Retrieval availability is typically held to a higher target than ingestion availability, since brief ingestion interruptions are often recoverable (source data can be buffered and re-submitted) while retrieval unavailability directly affects users. Implementers should set targets appropriate to their national operational context and infrastructure. Note that availability targets govern day-to-day operations; disaster recovery objectives (RTO/RPO) are addressed separately in /req/reliability/backup-recovery.

7.2.3.2. Data Integrity

Requirement 59	
Identifier	/req/reliability/data-integrity
Statement	The system SHALL ensure data integrity through: • ACID-compliant transactions for all data create, update, and delete operations • Referential integrity constraints preventing orphaned observations (observations without valid station references) • Checksums or cryptographic hash verification for file-based data transfers • Write-ahead logging or equivalent mechanisms to prevent data loss during system failures • Periodic integrity checks to detect and report data corruption

NOTE

Write-ahead logging (WAL) is a standard database durability mechanism in which changes are written to a transaction log before being applied to the main data store. In the event of a system crash, the log enables recovery to a consistent state. The specific mechanism is implementation-dependent; the requirement is that the system can recover to a consistent state following an unplanned failure.

Periodic integrity checks refer to automated verification that stored data matches expected checksums or hash values, detecting silent corruption caused by storage media failures or software defects. Results should be logged and administrators alerted on any detected inconsistency.

Referential integrity should be enforced as a system invariant. However, implementations receiving data from operational feeds (e.g., GTS, WIS 2.0) may encounter observations for stations not yet registered in the metadata repository. In such cases, implementations may auto-create a minimal station record pending metadata completion, rather than rejecting the observation. This exception should be logged and flagged for follow-up. See also /req/data-ingestion/validation.

7.2.3.3. Backup and Recovery

Requirement 60	
Identifier	/req/reliability/backup-recovery
Statement	The system SHALL implement backup and recovery capabilities: • Automated regular backups of all observation data, metadata, and system configuration • Defined and documented Recovery Time Objective (RTO) and Recovery Point Objective (RPO) appropriate to the operational context • Point-in-time recovery capability to restore the system to a known-good prior state • Documented and regularly tested recovery procedures • Verification of backup integrity through periodic restore testing

NOTE

RTO defines the maximum acceptable time to restore operations following a failure. RPO defines the maximum acceptable data loss measured in time. Implementers should define targets appropriate to their national context — for a national CDMS, a pragmatic RPO will often be longer than for a real-time operational system, since historical observation data is typically re-ingestible from source systems. The critical concern is protecting processed outputs (QC flags, homogenisation adjustments) that may not be reproducible without significant effort.

Backup storage volumes grow with retention period. Implementers should balance retention requirements against available storage, using incremental backup strategies and compression to manage costs. WMO-No. 1238 requires permanent retention of original and irreproducible climate records (3.4.1.4); backup policies should reflect this for long-term archival.

7.2.3.4. Fault Tolerance

Requirement 61	
Identifier	/req/reliability/fault-tolerance
Statement	The system SHALL handle failures gracefully by: • Continuing to serve read requests when write operations are unavailable • Queuing ingestion requests during temporary processing unavailability • Providing meaningful error messages rather than generic failures • Logging all error conditions with sufficient detail for diagnosis • Automatically retrying failed operations with exponential backoff

7.2.3.5. Data Validation on Retrieval

Requirement 62	
Identifier	/req/reliability/retrieval-validation
Statement	The system SHALL validate data consistency when retrieving stored data: • Verify checksums for file-based retrievals • Detect and report any data corruption to system administrators • Provide alternative access to backup copies when primary data is corrupted • Maintain audit logs of data integrity violations

NOTE	Corruption detection relies on comparing stored checksums or hash values against values recomputed at retrieval time. Alternative access to backup copies implies that backup data is accessible independently of the primary data store; this may be achieved through a secondary system, off-site backup, or disaster recovery capability (see /rec/reliability/disaster-recovery).

7.2.3.6. Disaster Recovery

Recommendation 10	
Identifier	/rec/reliability/disaster-recovery
Statement	The system SHOULD support disaster recovery capability including: • A secondary system or equivalent standby capable of assuming operations following catastrophic failure of the primary system • Documented switchover procedures, including manual activation steps and validation checks to confirm data integrity following switchover • Replication or backup transfer to a geographically separate location • Regular testing of disaster recovery procedures to verify recoverability

NOTE	A disaster recovery capability (sometimes termed a cold standby or Disaster Recovery Centre) provides resilience against catastrophic failure of the primary infrastructure (fire, flood, hardware failure). It is distinct from high-availability clustering, which addresses routine failures within the primary site. Switchover procedures should be documented in system maintenance documentation as required by /req/maintainability/documentation.

7.3. Requirements Class: Security

Requirements class	/req/security
Target type	Software system
Dependency	/req/core

7.3.1. Overview

Protect climate data assets, ensure appropriate access controls, maintain confidentiality where required, and provide comprehensive audit trails of security-relevant events. Security safeguards the integrity and

trustworthiness of the climate record while enabling appropriate access for legitimate users.

7.3.2. Justification and Regulatory Basis

WMO-No. 1131 requires "controlled access to data and systems" as a Required governance component (Section 3.2.1.1), specifying that "staff who have write access to data stored within a database only be able to change data under software control" with comprehensive audit trails. The specification mandates "each successful change to observations data be audited to ensure that the change, the details of the operator who made the change and the time of the change have all been recorded" (Section 3.2.1.1).

WMO-No. 1238 requires "Members shall ensure Entities implement rigorous authorization processes for modification or deletion of data or metadata" (3.4.4.1) and mandates "regularly review systems for the physical- and cyber-security of the climate record" (3.4.4.2).

7.3.3. Requirements

7.3.3.1. Authentication

Requirement 63	
Identifier	/req/security/authentication
Statement	The system SHALL implement authentication mechanisms that: • Support integration with institutional or enterprise identity management systems using standard authentication protocols • Enforce strong credential policies for locally managed accounts • Support multi-factor authentication (MFA) for administrative and privileged accounts • Implement account lockout after repeated failed authentication attempts • Maintain secure session management with automatic timeout after inactivity

NOTE	Standard authentication protocols include OpenID Connect, OAuth 2.0, SAML 2.0, and LDAP, among others. The choice of protocol should reflect the identity management infrastructure in use within the deploying organisation. OpenID Connect is widely adopted and provides a standards-based federated identity layer suitable for both web and API access.

7.3.3.2. Authorization and Access Control

Requirement 64	
Identifier	/req/security/authorization

Statement	The system SHALL implement role-based access control (RBAC) that: • Separates privileges across user roles including at minimum: read-only users, data entry operators, quality control analysts, and system administrators • Is configurable to restrict data access at station or network level • Enforces data access policies for embargoed or sensitive datasets • Applies the principle of least privilege: users are granted the minimum permissions necessary for their role • Allows system administrators to define and manage custom roles and their associated permissions

NOTE	Role definitions and their associated permissions should be configurable by system administrators without requiring code changes or system restart. Access restrictions at station or network level are important for national systems where data from certain networks may be subject to bilateral sharing agreements or national security constraints.

7.3.3.3. Data Encryption

Requirement 65	
Identifier	/req/security/encryption
Statement	The system SHALL implement encryption for: • All data in transit using current industry-standard transport security protocols • Sensitive data at rest including user credentials, API keys, and personal information • Backup files stored on external or off-site media

7.3.3.4. Audit Logging

Requirement 66	
Identifier	/req/security/audit-logging
Statement	The system SHALL:
A	maintain comprehensive audit logs recording: • All user authentication attempts (successful and failed) • All data and metadata modifications (create, update, delete) with user identification • All administrative operations (configuration changes, user management, permission changes) • All quality control flag overrides with justification • All data export operations including user, timestamp, and data selection criteria

B	Audit logs SHALL: • Be tamper-evident or stored in write-once media • Be retained in accordance with the organisation's data retention policy • Be searchable and exportable for security incident investigation • Include sufficient detail to reconstruct the sequence of events

7.3.3.5. Vulnerability Management

Requirement 67	
Identifier	/req/security/vulnerability-management
A	The system SHALL support security maintenance through: • Regular security updates to address known vulnerabilities • Documented process for applying security patches • Vulnerability scanning capability or support for external security assessment tools • Security advisories for known issues affecting the deployment

7.3.3.6. Data Privacy

Requirement 68	
Identifier	/req/security/data-privacy
A	The system SHALL protect personal information by: • Minimising collection and storage of personally identifiable information (PII) to what is operationally necessary • Providing mechanisms to redact or anonymise personal information where required • Supporting compliance with applicable data protection regulations as determined by the deploying organisation • Documenting data retention policies for any personal data held by the system

NOTE	The applicable data protection regulations will vary by jurisdiction. Implementers should assess which regulations apply to their deployment (e.g., national privacy laws, regional frameworks) and configure the system accordingly. The primary personal data held by a CDMS is typically contact information for station operators and data submitters rather than observation data itself.

7.3.3.7. Provenance Security

Requirement 69	
Identifier	/req/security/provenance-security

A	The system SHALL protect provenance records through: • Cryptographic timestamps or equivalent mechanisms to prevent backdating • Tamper-evident storage for provenance audit trails

7.4. Requirements Class: Usability

Requirements class	/req/usability
Target type	Software system
Dependency	/req/core

7.4.1. Overview

Ensure the system is accessible, intuitive, and efficient for climate scientists, data managers, and operators with varying levels of technical expertise. Usability directly impacts the effectiveness of quality control, data validation, and metadata management workflows that depend on expert human judgment.

7.4.2. Justification and Regulatory Basis

WMO-No. 1131 emphasizes that CDMSs must support diverse user roles including, *inter alia*, "climate scientists, data managers, system administrators, and data entry operators" (Figure 2 and Figure 3). The specification notes that while "most NMHSs would not manage infrastructure and software required to support" certain advanced components, systems must still be " flexible enough to accommodate national climate monitoring products and other national indices" (Section 4.4.2.3).

WMO-No. 1238 recognizes the need for "climate scientists to review and validate climate observations, identify inhomogeneities" (Figure 2), which requires effective user interfaces. The Competencies requirements (3.6) acknowledge that staff need appropriate training, implying systems should support learnable, efficient workflows.

7.4.3. Requirements

7.4.3.1. User Interface Design

Requirement 70	
Identifier	/req/usability/ui-design
A	The system user interface SHALL: • Provide consistent navigation patterns across all functions • Use clear, unambiguous labels and terminology aligned with climate science conventions • Provide contextual help and documentation accessible from relevant screens • Support keyboard navigation for efficiency • Indicate required fields and validation requirements clearly before submission • Provide meaningful error messages with guidance on corrective actions

NOTE	A web-based user interface is the recommended approach as it maximises accessibility across platforms, reduces client-side deployment complexity, and facilitates integration with

web-based identity management systems. Implementations providing a web-based interface should consider the accessibility recommendations in /rec/usability/accessibility.

7.4.3.2. Accessibility

Recommendation 11	
Identifier	/rec/usability/accessibility
A	The system user interface SHOULD conform to: • Web Content Accessibility Guidelines (WCAG) 2.1 Level AA where applicable • Support for screen readers and assistive technologies • Sufficient color contrast for visual accessibility • Text scaling without loss of functionality • Keyboard-only operation for all critical functions

7.4.3.3. User Feedback and Notifications

Requirement 71	
Identifier	/req/usability/notifications
A	The system SHALL provide notifications for: • Completion of long-running operations (ingestion, QC processing, homogenization) • Critical quality control failures requiring manual review • System maintenance schedules • Data embargo or access restriction changes • Configurable notification delivery methods (email, in-application alerts)

7.5. Requirements Class: Maintainability

Requirements class	/req/maintainability
Target type	Software system
Dependency	/req/core

7.5.1. Overview

Ensure the system can be updated, configured, and troubleshot efficiently with minimal disruption to operations. Maintainability is critical for long-term sustainability, as climate data systems must operate reliably over multiple decades with evolving technologies and requirements.

7.5.2. Justification and Regulatory Basis

WMO-No. 1131 emphasizes that "one of the steps often overlooked when implementing a CDMS is to ensure that the NMHS has established a solid governance framework to protect the integrity of the climate record" (Section 3), including "processes that ensure adherence to CDMS-related policies" and

comprehensive documentation requirements (Section 3.2.2.5). The specification requires "effective IT security and monitoring" (Section 5.2.3) and "managed change" as Required governance (Section 3.2.2.2), noting that "uncontrolled change can result in chaos for users" including "data corruptions or even data loss."

WMO-No. 1238 mandates "Members shall ensure that Entities communicate work involving changes to the data management system and associated software to affected stakeholders in a timely fashion" (3.3.3.3) and requires comprehensive documentation of "data management processes, including details of roles and responsibilities" (3.3.3.1).

7.5.3. Requirements

7.5.3.1. Configuration Management

Requirement 72	
Identifier	/req/maintainability/configuration
A	The system SHALL support configuration management with: • Externalized configuration files separate from application code • Hot-reload capability for quality control threshold updates without system restart • Configuration version control with rollback capability • Configuration validation before activation to prevent invalid settings • Documentation of all configurable parameters with valid value ranges

7.5.3.2. Logging and Diagnostics

Requirement 73	
Identifier	/req/maintainability/logging
A	The system SHALL provide logging and diagnostic capabilities: • Configurable log levels (ERROR, WARN, INFO, DEBUG) per component • Structured logging with timestamps, component identification, and correlation IDs • Log rotation and retention policies to manage disk space • Centralized logging capability for distributed deployments • Performance metrics logging (response times, throughput, resource utilization)

7.5.3.3. Health Monitoring

Requirement 74	
Identifier	/req/maintainability/health-monitoring

A	The system SHALL expose health monitoring endpoints providing: • Overall system health status (healthy, degraded, unhealthy) • Component-level health status for all major system components • Key performance indicators (query response times, ingestion rates, error rates) • Resource utilization metrics (CPU, memory, disk, network) • Integration with standard monitoring protocols or tools used within the deployment environment

7.5.3.4. API Backward Compatibility

Requirement 75	
Identifier	/req/maintainability/api-compatibility
A	The system SHALL maintain API stability through: • Semantic versioning of API endpoints • A defined and published deprecation policy specifying the support period for deprecated API versions • Clear deprecation notices in API documentation and response headers • Migration guides for breaking changes • Ability to run multiple API versions concurrently during transition periods

7.5.3.5. Database Schema Evolution

Requirement 76	
Identifier	/req/maintainability/schema-evolution
A	The system SHALL support database schema changes with: • Version-controlled schema migration scripts • Automated or documented migration procedures • Ability to test schema migrations in non-production environments • Rollback capability for failed migrations • Backward compatibility for schema changes where feasible

7.5.3.6. Documentation

Requirement 77	
Identifier	/req/maintainability/documentation

A	The system SHALL provide comprehensive documentation including: • System architecture and component descriptions • Installation and deployment procedures • Configuration reference with parameter descriptions and examples • API reference documentation with request/response examples • Troubleshooting guides for common issues • Backup and recovery procedures • Disaster recovery and system switchover procedures • Security hardening guidelines

7.6. Requirements Class: Interoperability

Requirements class	/req/interoperability
Target type	Software system
Dependency	/req/core, /req/data-api

7.6.1. Overview

Ensure the system can exchange data and integrate with external systems using industry-standard protocols, formats, and conventions. Interoperability enables climate data to flow seamlessly between national systems, regional centers, international archives, and the WIS 2.0 infrastructure.

7.6.2. Justification and Regulatory Basis

WMO-No. 1131 emphasizes that "open spatial standards are a mechanism that is being increasingly adopted by many organizations and industry sectors around the world" (Section 8.1) and that these standards are "expected to become an increasingly important mechanism for distributing data in future years." The specification notes that standardized delivery "will ensure that data are represented in a form that facilitates data interoperability and are accessible to a wide range of end users from disparate industries" (Section 3.1.4.1).

WMO-No. 1238 requires "Members shall ensure that Entities encode data, and associated metadata, intended for international exchange following the standards specified in the Manual on Codes (WMO-No. 306), the Manual on the WMO Information System (WMO-No. 1060)" (3.2.2.1). The framework principle mandating registration in "the WIS Global Discovery Catalogue" (1.1.1(4)) requires standards-based interoperability.

7.6.3. Requirements

7.6.3.1. Standards Compliance

Requirement 78	
Identifier	/req/interoperability/standards

A	The system SHALL comply with relevant international standards: • WMO WIGOS Metadata Standard for station metadata • WMO Manual on Codes (WMO-No. 306) for BUFR and code form support • CF Conventions for NetCDF climate data files • OGC standards (API - Features, API - EDR, API - Records, or SensorThings API) for data access and discovery APIs

7.6.3.2. API Interoperability

Requirement 79	
Identifier	/req/interoperability/api
A	The system API SHALL support interoperability through: • Content negotiation using standard HTTP Accept headers • Standard HTTP status codes with appropriate semantics • CORS (Cross-Origin Resource Sharing) support for web-based clients • OpenAPI (Swagger) specification or equivalent machine-readable API description

7.7. Requirements Class: Data Governance

Requirements class	/req/data-governance
Target type	Software system
Dependency	/req/core

7.7.1. Overview

Ensure proper stewardship of climate data through lifecycle management, quality documentation, access policies, and compliance with organizational and regulatory requirements. Data governance provides the policy framework and accountability structures necessary for long-term climate data stewardship.

7.7.2. Justification and Regulatory Basis

WMO-No. 1131 dedicates Chapter 3 entirely to "CDMS Governance", stating that "one of the steps often overlooked when implementing a CDMS is to ensure that the NMHS has established a solid governance framework to protect the integrity of the climate record" involving "Policies" and "Governance processes". The specification defines data governance as Required (Section 3.2.1), including "controlled access to data and systems", "approval process for new data types", "approval process to change data", and "IT change approvals".

WMO-No. 1238 requires "Members should ensure that Entities implement and maintain a governance mechanism to oversee the various governance elements, coordinated by a governance body" (3.5.1.1) and mandates policies covering commitments, sustainability, intellectual property, and data delivery (Section 3.1 of WMO-No. 1131). The framework emphasizes that "data stewardship assigns to an entity rights and responsibilities for acquiring and managing climate data and information" (Definitions), requiring formalized governance structures.

7.7.3. Requirements

7.7.3.1. Data Lifecycle Management

Requirement 80	
Identifier	/req/data-governance/lifecycle
Statement	The system SHALL support data lifecycle management with: • Configurable data retention policies per dataset type or network • Automated archival of historical data to long-term storage • Data deletion workflows with approval requirements for permanent deletion • Tombstone records for deleted data maintaining metadata about what was deleted and why • Migration paths for data format updates or schema changes

7.7.3.2. Data Quality Metadata

Requirement 81	
Identifier	/req/data-governance/quality-metadata
Statement	The system SHALL maintain data quality metadata including: • Completeness metrics (percentage of expected observations received) • Timeliness metrics (delay between observation time and system receipt) • Quality control statistics (pass/fail rates per check type) • Manual intervention frequency and types • Homogenization status and adjustment statistics • Overall fitness-for-purpose assessments
B	Quality metadata SHALL be: • Queryable alongside observation data • Exportable in machine-readable formats • Updated automatically as data quality changes

NOTE	Fitness-for-purpose refers to an assessment of whether a dataset meets the quality requirements of a particular application. For example, a dataset with frequent short gaps may be suitable for climatological trend analysis but unsuitable for computing climate normals. Fitness-for-purpose assessments summarise quality metrics (completeness, temporal continuity, QC flag distribution) in terms of suitability for defined use cases, supporting informed data selection by users. See also /req/metadata-management/data-quality.

7.7.3.3. Data Access Policies

Requirement 82	
Identifier	/req/data-governance/access-policies

Statement	The system SHALL enforce data access policies including: • Public/restricted/confidential data classification • Embargo periods for recently collected data where required • License and attribution requirements for data usage • Data sharing agreements and restrictions per network or organization • Automated enforcement of access restrictions based on user roles

7.7.3.4. Data Stewardship Roles

Requirement 83	
Identifier	/req/data-governance/stewardship
Statement	The system SHALL define and support data stewardship roles including: • Data custodian responsibilities for dataset maintenance • Data quality manager authority for quality control decisions • Data access coordinator for managing access requests and policies • System administrator separation from data stewardship roles • Clear escalation paths for data governance issues

NOTE	Data stewardship roles define the organisational responsibilities for data management and are distinct from the technical access control roles defined in /req/security/authorization. In practice, stewardship roles map to RBAC roles: for example, the data quality manager role is associated with permissions to override quality control flags and approve homogenisation adjustments. The RBAC configuration in the security class provides the technical enforcement of the responsibilities defined here.